Kubernetes Static Website Deployment

cloud-devops-multicloud-AWS-AZURE-python-ai-projects/06-kubernetes$ tree -a

```
.
└── 01-kubernetes-static-website
    └── k8s
2 directories, 0 files
```

project structure

```
cloud-devops-multicloud-AWS-AZURE-python-ai-projects/
|
├── 05-docker-containers/
|   └── 01-dockerized-static-website/
|       ├── Dockerfile
|       ├── .dockerignore
|       ├── README.md
|       ├── index.html
|       └── css/
|           └── style.css
|
├── 06-kubernetes/
|   ├── README.md
|   |
|   └── 01-kubernetes-static-website/
|       ├── README.md
|       └── k8s/
|           ├── namespace.yaml
|           ├── deployment.yaml
```

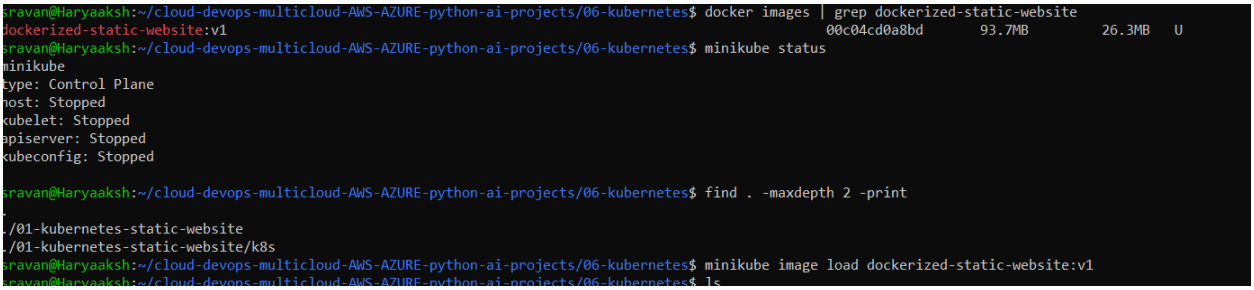
| └── service.yaml

|

└── README.md

We'll use your existing Docker image: dockerized-static-website:v1

And load it into minikube before deployment:
minikube image load dockerized-static-website:v1



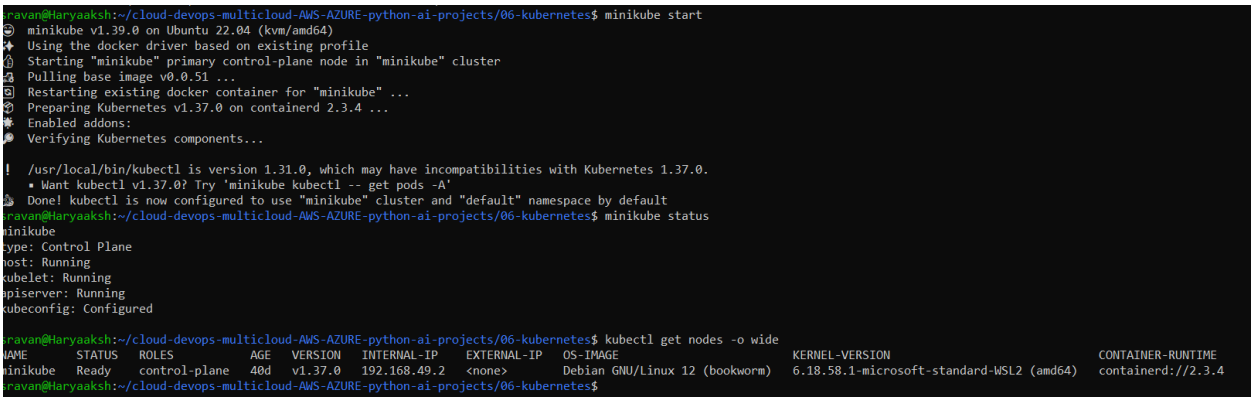
We will first verify that your local Docker image exists and that Minikube is running before applying any Kubernetes manifests.

We will not copy the website's HTML/CSS or create another Dockerfile in the Kubernetes directory. Kubernetes will reuse the existing image.

Phase 1 — Start minikube

Step 1: Start your Kubernetes cluster

minikube start
minikube status
kubectl get nodes –o wide



Phase 2 — Load your Docker image

Once Minikube is running, execute:

minikube image load dockerized-static-website:v1

Verify the image is available inside Minikube:
minikube image ls | grep dockerized-static-website

docker.io/library/dockerized-static-website:v1

Phase 3 — Create a Kubernetes Namespace

First, let's understand what a Namespace is.

A Kubernetes Namespace logically separates resources inside a cluster. For example, you could have separate namespaces for development, testing and production.

For our project, we'll create:

Namespace: static-website

All the Kubernetes resources for this project will be deployed inside this namespace.

vi namespace.yaml

cat namespace.yaml

apiVersion: v1

kind: Namespace

metadata:

name: static-website

labels:

app: static-website

environement: learning

```
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/06-kubernetes/01-kubernetes-static-website/k8s$ cat namespace.yaml
apiVersion: v1
kind: Namespace
metadata:
  name: static-website
  labels:
    app: static-website
    environement: learning

sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/06-kubernetes/01-kubernetes-static-website/k8s$ kubectl get --namespace
error: flag needs an argument: --namespace
See 'kubectl get --help' for usage.
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/06-kubernetes/01-kubernetes-static-website/k8s$ kubectl get --namespaces
error: unknown flag: --namespaces
See 'kubectl get --help' for usage.
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/06-kubernetes/01-kubernetes-static-website/k8s$ kubectl get --namespace=static-website
You must specify the type of resource to get. Use "kubectl api-resources" for a complete list of supported resources.

error: Required resource not specified.
Use "kubectl explain <resource>" for a detailed description of that resource (e.g. kubectl explain pods).
See 'kubectl get -h' for help and examples
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/06-kubernetes/01-kubernetes-static-website/k8s$ minikube kubectl -- get namespaces
    > kubectl.sha256:  64 B / 64 B [-------------------------] 100.00% ? p/s 0s
    > kubectl:  59.02 MiB / 59.02 MiB [--------------] 100.00% 1.24 MiB p/s 48s
NAME              STATUS   AGE
default           Active   40d
dev-pod           Active   7d9h
kube-node-lease   Active   40d
kube-public       Active   40d
kube-system       Active   40d
static-website    Active   117s
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/06-kubernetes/01-kubernetes-static-website/k8s$ kubectl -- get namespaces
```

minikube kubectl -- get namespaces

NAME STATUS AGE

default Active 40d

dev-pod Active 7d9h

kube-node-lease Active 40d

kube-public Active 40d

kube-system Active 40d

static-website Active 117s

Phase 4 — Create Kubernetes Deployment

1. Understand the architecture

We currently have:

- Minikube Kubernetes cluster running.
- Namespace: static-website.
- Docker image: dockerized-static-website:v1.
- Nginx web server inside that image.

We want Kubernetes to maintain three running Pods.

```
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/06-kubernetes/01-kubernetes-static-website/k8s$ cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: static-website-deployment
  namespace: static-website
  labels:
    app: static-website
spec:
  replicas: 3
  selector:
    matchLabels:
      app: static-website
  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxSurge: 1
      maxUnavailable: 0
  template:
    metadata:
      labels:
        app: static-website
    spec:
      containers:
        - name: nginx-website
          image: dockerized-static-website:v1
          imagePullPolicy: Never
          ports:
            - containerPort: 80
              name: http
          resources:
            requests:
              cpu: "50m"
              memory: "64Mi"
            limits:
              cpu: "250m"
              memory: "128Mi"

sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/06-kubernetes/01-kubernetes-static-website/k8s$ cat namespace.yaml
apiVersion: v1
kind: Namespace
metadata:
  name: static-website
  labels:
    app: static-website
    environement: learning
```

```
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/06-kubernetes/01-kubernetes-static-website/k8s$ kubectl apply -f deployment.yaml
deployment.apps/static-website-deployment created
```

```
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/06-kubernetes/01-kubernetes-static-website/k8s$ kubectl rollout status deployment/static-website-deployment -n static-website
deployment "static-website-deployment" successfully rolled out
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/06-kubernetes/01-kubernetes-static-website/k8s$ kubectl  get deployments -n static-website
NAME                     READY   UP-TO-DATE   AVAILABLE   AGE
static-website-deployment   3/3     3            3           91s
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/06-kubernetes/01-kubernetes-static-website/k8s$ kubectl get replicasets -n static-website
NAME                               DESIRED   CURRENT   READY   AGE
static-website-deployment-74fcc4fdf9   3         3         3       110s
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/06-kubernetes/01-kubernetes-static-website/k8s$ kubectl get pods -n static-website -o wide
NAME                                     READY   STATUS    RESTARTS   AGE    IP            NODE       NOMINATED NODE   READINESS GATES
static-website-deployment-74fcc4fdf9-9q7cz   1/1     Running   0          2m8s   10.244.0.25   minikube   <none>           <none>
static-website-deployment-74fcc4fdf9-cb2mz   1/1     Running   0          2m8s   10.244.0.24   minikube   <none>           <none>
static-website-deployment-74fcc4fdf9-k27m4   1/1     Running   0          2m8s   10.244.0.23   minikube   <none>           <none>
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/06-kubernetes/01-kubernetes-static-website/k8s$
```

kubectl apply -f deployment.yaml

deployment.apps/static-website-deployment created

kubectl rollout status deployment/static-website-deployment -n static-website

deployment "static-website-deployment" successfully rolled out

kubectl get deployments -n static-website

NAME   READY   UP-TO-DATE   AVAILABLE   AGE

static-website-deployment   3/3   3   3   91s

kubectl get replicasets -n static-website

NAME   DESIRED   CURRENT   READY   AGE

static-website-deployment-74fcc4fdf9   3   3   3   110s

kubectl get pods -n static-website -o wide

NAME   READY   STATUS   RESTARTS   AGE   IP   NODE
NOMINATED NODE   READINESS GATES

static-website-deployment-74fcc4fdf9-9q7cz   1/1   Running   0   2m8s   10.244.0.25
minikube   <none>   <none>

static-website-deployment-74fcc4fdf9-cb2mz   1/1   Running   0   2m8s   10.244.0.24
minikube   <none>   <none>

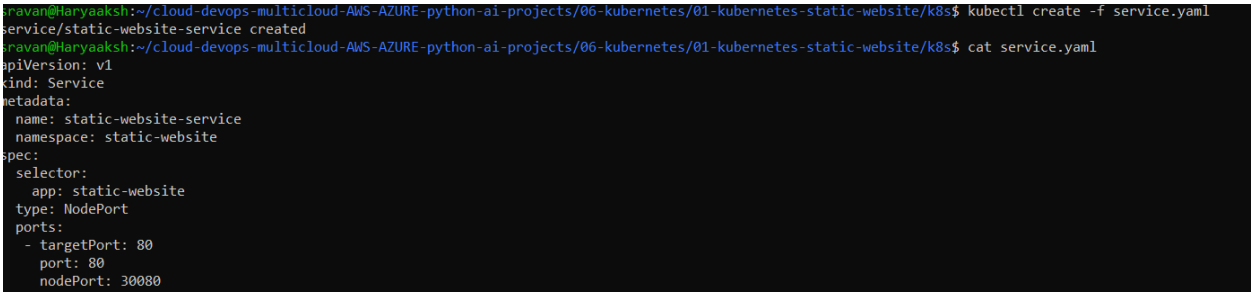
static-website-deployment-74fcc4fdf9-k27m4   1/1   Running   0   2m8s   10.244.0.23
minikube   <none>   <none>

Phase 5 — Create Kubernetes Service

Now we need to expose website so that users can access it.

Currently, the three Pods have internal IP addresses. We don't want users to access individual Pods directly because Pod IPs can change.

We'll create a NodePort Service.



vi service.yaml

sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/06-kubernetes/01-kubernetes-static-website/k8s$ kubectl create -f service.yaml

service/static-website-service created

sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects/06-kubernetes/01-kubernetes-static-website/k8s$ cat service.yaml

apiVersion: v1

kind: Service

metadata:

name: static-website-service

namespace: static-website

spec:

selector:

app: static-website

type: NodePort

ports:

- targetPort: 80

port: 80

kubectl get services -n static-website

NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE

static-website-service NodePort 10.96.201.177 <none> 80:30080/TCP 84s

Verify the Service and endpoints
kubectl get endpoints -n static-website

Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice

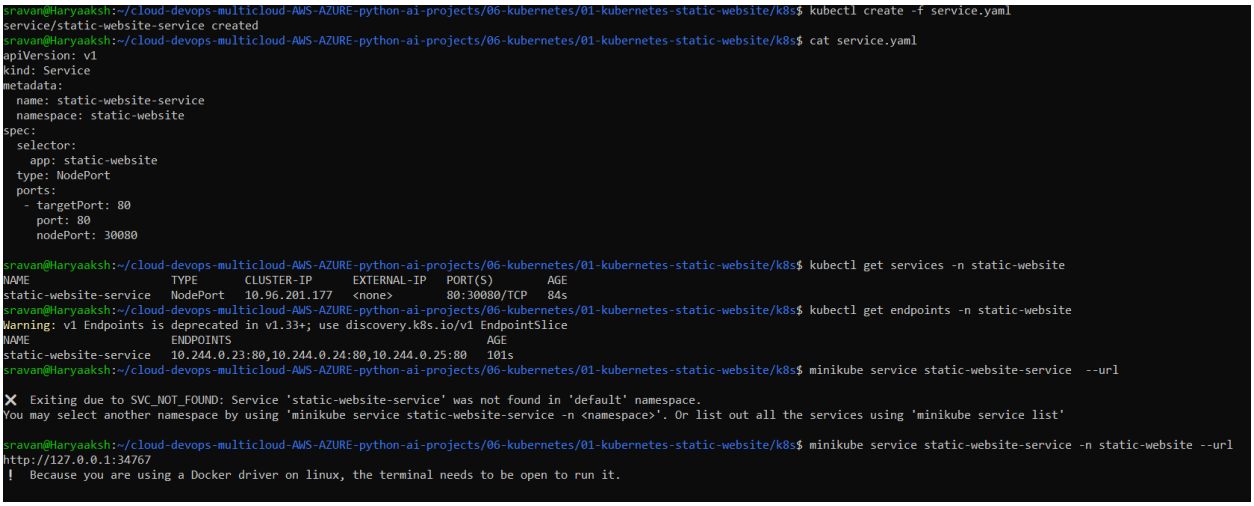
NAME ENDPOINTS AGE

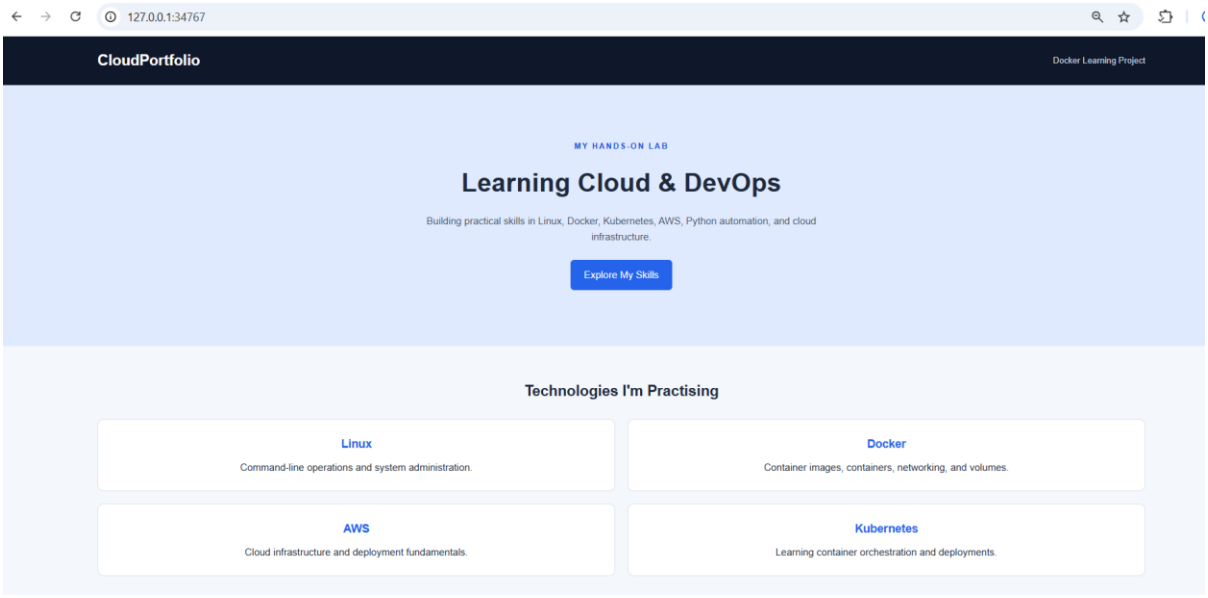
static-website-service 10.244.0.23:80,10.244.0.24:80,10.244.0.25:80 101s

Access website
minikube service static-website-service -n static-website --url

http://127.0.0.1:34767

! Because you are using a Docker driver on linux, the terminal needs to be open to run it.





cloud-devops-multicloud-AWS-AZURE-python-ai-projects$
git add 06-kubernetes/01-kubernetes-static-website/

```
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects$ git add 06-kubernetes/01-kubernetes-static-website/
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects$ git diff --cached --name-status
A       06-kubernetes/01-kubernetes-static-website/README.md
A       06-kubernetes/01-kubernetes-static-website/k8s/deployment.yaml
A       06-kubernetes/01-kubernetes-static-website/k8s/namespace.yaml
A       06-kubernetes/01-kubernetes-static-website/k8s/service.yaml
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects$ git commit -m "feat: deploy static website on Kubernetes using Minikube"
[main ee37594] feat: deploy static website on Kubernetes using Minikube
 4 files changed, 327 insertions(+)
 create mode 100644 06-kubernetes/01-kubernetes-static-website/README.md
 create mode 100644 06-kubernetes/01-kubernetes-static-website/k8s/deployment.yaml
 create mode 100644 06-kubernetes/01-kubernetes-static-website/k8s/namespace.yaml
 create mode 100644 06-kubernetes/01-kubernetes-static-website/k8s/service.yaml
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects$ git push origin main
Username for 'https://github.com': sravanboosa
Password for 'https://sravanboosa@github.com':
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 8 threads
Compressing objects: 100% (9/9), done.
Writing objects: 100% (9/9), 3.08 KiB | 3.08 MiB/s, done.
Total 9 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/SravanBoosa/cloud-devops-multicloud-AWS-AZURE-python-ai-projects.git
   28e34cb..ee37594  main -> main
sravan@Haryaaksh:~/cloud-devops-multicloud-AWS-AZURE-python-ai-projects$
```